



# Numerical Analysis of Coaxially 3D Printed Lunar Habitats: Integrating Regolith and PCM for Passive Temperature Control

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## Abstract

Space research is shifting its focus from low-gravity platforms to Moon and Mars exploration, requiring advanced habitat construction. This paper proposes to integrate the 3D printing of regolith and Phase Change Materials (PCM), with a particular interest in lunar habitats. A coaxial printing approach is numerically analyzed, enabling the simultaneous deposition of a regolith shell, providing structural integrity, and a PCM core that helps regulate the interior habitat temperature in a passive manner. Simulations show that 3D printing with a coaxial nozzle can effectively control the shell-core (internal) structure of the habitat wall by adjusting printing parameters or by constructing multi-strand composite walls. The PCM core becomes thicker when using a smaller layer height or a higher extrusion speed during printing. We then examine the expected thermal response of the habitat, initially made of pure regolith and later incorporating the PCM. With just regolith, results indicate that one can select adequate thermo-optical properties at the wall external boundary, and wall thickness, to control the mean temperature in the interior, and associated fluctuations, respectively. Incorporating the PCM, either in single or multiple regolith-PCM strands, is shown to effectively stabilize this internal temperature ( $\Delta T_i \rightarrow 0$  K), improving thermal performance and habitat design, and reducing the quantity of required regolith and binder. Since binder has to be brought from Earth, this reduction can be a major factor in enabling sustainable construction on the Moon and beyond.

**Keywords** Space construction · Thermal management · 3D printing · Shell-core technique · PCM · Lunar regolith

## Introduction

As space science enters a new era, the focus is shifting from microgravity platforms, like the International Space Station, to celestial bodies, primarily the Moon — as it can serve as an intermediate point for other extraterrestrial missions, with a special focus on its South Pole (Schunk et al. 2022) — and Mars (Salgado-Rodríguez et al. 2022; Lee et al. 2022). Lunar and martian missions represent the human return to extraterrestrial travel and will require encompassing both fundamental and applied science (Vailati et al. 2024). The access to scarce natural resources on Earth and unique

conditions for scientific research, and the potential for manufacturing advanced materials are just a few attractive reasons to venture beyond Earth's manned spaceflight.

Efforts to establish a permanent human presence on the Moon (e.g., the ESA Moon Village program (ESA 2022)) focus on designing and constructing habitats that protect astronauts from environmental hazards, including the absence of atmosphere, radiation exposure, and meteorite impacts (Ulubeyli 2022). The diverse and harsh conditions that can be experienced on the lunar surface will significantly influence such design and other aspects of the mission. For instance, it is well known that the Moon's surface exhibits extreme temperature variations between  $-244$  and  $122$  °C, depending on the geographical coordinates and associated eclipse duration, landscape shadows, and local composition of the surface regolith. For the sustained presence of astronauts, the habitat should be capable of insulating the crew from these extreme conditions and provide a suitable, living environment in its interior, which entails the design of an effective thermal control system. In this context, we propose the 3D printing of habitat structures that integrate regolith,

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providing structural integrity, and Phase Change Materials (PCM) for passive temperature control.

Due to their ability to exchange large amounts of heat (at nearly constant temperature) during their phase transition, PCM have long history of use for thermal regulation in many applications (Mondal 2008; Mehling 2023; Khan et al. 2017; Garmendia et al. 2022), including space-relevant ones (Humphries and Griggs 1977; Khan and Singh 2024; Diaconu et al. 2024; Peng et al. 2022). Organic PCM, in particular, have gained growing attention in the space sector due to their large heat of fusion, chemical stability and compatibility with most materials, and moderate melting temperatures. Their practical use, however, has been somewhat limited given their low thermal conductivity, which reduces responsiveness to thermal variations. On Earth, this limitation is partially mitigated by natural convection, providing a natural source of convective transport. On the Moon, characterized by approximately one-sixth of Earth's gravity, or other microgravity environments, the absence of strong convection would pose a significant challenge, reducing PCM performance by a significant factor.

Recently, the concept of thermocapillary-enhanced PCM (Salgado Sánchez et al. 2020; Ezquerro et al. 2020) has been proposed — and recognized by ESA — as a promising alternative to improve PCM performance in reduced gravity. Thermocapillary convection arises from a surface tension gradient along a liquid PCM-air interface, caused by temperature variations. This gradient generates a tangential force that drives fluid motion close to the interface. In this context, the MarPCM project (Porter et al. 2023; Martínez et al. 2024) aims to explore various thermocapillary-based strategies, including different PCM geometries (Varas et al. 2021; Borshchak Kachalov et al. 2024b; Šeta et al. 2021), to enhance heat transport in PCM devices that incorporate a free surface. Although experiments are scheduled to be performed on the ISS no earlier than 2026 (schedule permitting), numerical results have already supported the use of the thermocapillary effect, either alone (Varas et al. 2021; Salgado Sanchez et al. 2020; Borshchak Kachalov et al. 2024a) or combined with (weak) gravitational convection (Borshchak Kachalov et al. 2024b, 2021; Varas et al. 2023), to improve PCM performance. Related research has also looked at supplemental strategies based on geometry (Borshchak Kachalov et al. 2024b, 2022), material (Peng et al. 2021, 2025), dual-PCM (Kansara et al. 2025) or multi-cycle optimization (Dubert et al. 2024).

PCM can thus offer a passive solution for stabilizing temperatures in lunar habitats, absorbing excess heat during the lunar day and releasing it at night. This approach reduces reliance on active thermal control systems, conserving energy and extending the operational lifespan of critical mission components. Integrating PCM into regolith-based structures is expected to enhance insulation and overall

thermal performance, similar to the strategies used in net-zero energy buildings (NZEBs) on Earth (Tyagi and Buddhi 2007; Parameshwaran et al. 2010).

A recent preliminary study explored the suitability of PCM as thermal control devices for space habitats, and provided guidelines for selecting key design parameters, such as the thermo-optical properties of the external habitat boundary controlling the absorption of solar radiation (Borshchak Kachalov et al. 2023). More recently, scaled laboratory experiments on semi-spherical (igloo-shaped) habitats confirmed these findings. These ground tests subjected the habitat to repeated thermal cycles that simulate illumination (melting) and eclipse (solidification) cycles (phases), and demonstrated that the internal habitat temperature was significantly stabilized around the melting temperature of the PCM (Ongil et al. 2024). The present work aims to extend this seminal research, proposing the 3D printing of integrated regolith-PCM layers to construct the habitat structure.

3D printing, particularly material extrusion additive manufacturing, has been identified as a potential construction methodology for lunar habitats due to its ability to utilize local materials and adapt to in-situ conditions. On Earth, entire houses are 3D printed nowadays within short timeframes (UTU 3D 2025). On the Moon, 3D printing combined with In-Situ Resource Utilization (ISRU) can significantly reduce the need for transporting construction materials, lowering costs and logistical challenges. Regarding printing materials, lunar regolith is an abundant and readily available resource that can be harnessed as a primary material and, when combined with the 3D printing technique, it can be used to construct robust and durable structural shells. In terms of manufacturing methodology, the geopolymerization process is the most widely used, see, e.g., Ma et al. (2023). This consists of the extrusion of a paste made of a ceramic-like powder (regolith) (Cesarano 1998) and a binder — phosphoric acid (Buchner et al. 2018), alkaline agent (Momi et al. 2021a) or bio-organic compounds (Grundström et al. 2021) — that is cured at the end. In this regard, note that regolith simulants have been developed and used extensively for studies (see, e.g., Toklu and Akpınar 2022; Momi et al. 2021b and references therein), given the reduced availability of samples (Schreiner et al. 2016).

Such off-Earth manufacturing of structures heavily relies on insights gained from terrestrial applications. In this context, Computational Fluid Dynamics (CFD) modeling has proven valuable for optimizing key aspects of the 3D printing process, such as understanding the morphology of the deposited layers (Spangenberg et al. 2021; Xu et al. 2022) and internal shell-core structure (Narei et al. 2021), ensuring layer stability (Mollah et al. 2021, 2022), defining reinforcement strategies with rebars (Spangenberg et al. 2022; Mollah et al. 2023b) and fibers (Šeta et al. 2023, 2024), and controlling the curing of the material (Mollah et al. 2023b), among

other considerations. Mollah et al. (2025) recently explained the deposition of layers during off-Earth printing using typical cylindrical nozzles, and focusing on the curing of a slurry based on lunar regolith.

However, to effectively integrate PCM into multi-material printing, the use of coaxial nozzles must be investigated. Coaxial 3D printing allows for the simultaneous deposition of multiple materials in a layered configuration, making it possible to embed PCM, as a thermal-regulating core, within a structurally supportive regolith shell; this method is similar to incorporating PCM into concrete (Ates et al. 2024; Gencel et al. 2022) or geopolymers (Rahemipoor et al. 2023, 2024) but applied on a larger scale. This coaxial printing technology has already found a number of applications that benefit from it, e.g., vascular network applications in biomedicine, sensors and actuators, alloys with liquid metals, textile industry, and mechanical reinforcement (Rafiee et al. 2021; Kjar et al. 2021).

Here, we extend the application of PCM and coaxial 3D printing to the construction of space habitats by introducing a novel approach that simultaneously addresses passive thermal control and manufacturing methodology. Specifically, this paper presents the innovative concept of coaxially 3D printing lunar habitats using regolith and phase change materials, supported by combining simulations of the printing process and a comprehensive numerical analysis of the thermal performance of the resulting regolith-PCM structure. The manuscript is organized as follows. First, numerical models for both additive manufacturing and thermal analyses are described in Section “Methodology”. In Section “3D Printing Results”, results for the regolith-PCM coaxial 3D printing are presented and discussed for different representative printing conditions. Section “Thermal Analysis” then describes the results associated with the thermal analysis of the printed structures. These are presented for representative cases with and without PCM, and analyzed from a design perspective. Finally, Section “Conclusion” offers the main conclusions of the manuscript, including remarks about the feasibility of this habitat construction concept.

## Methodology

### 3D Printing Process

#### Materials and Rheology

Lunar regolith is the primary material when it comes to 3D printing of structures on the Moon. Unlike traditional building materials for on-Earth construction, lunar regolith possesses unique flow properties due to its fine-grained dust and rocky fragments. Factors such as particle size

distribution, shape, and surface roughness significantly influence its rheological behavior. Due to the limited access to lunar regolith, simulants that replicate the chemical and physical properties of the actual regolith are typically used for research. This study focuses on the simulant JSC-1A, which is produced through the milling and sieving of basalt cinders sourced from the Merriam Crater in Arizona (noa 2007). The JSC-1A simulant exhibits a chemical composition that includes elemental oxides compatible with those found in samples extracted in the Apollo missions (Lindsay 1976).

To prepare a 3D printable paste, lunar regolith must be combined with a binding agent. Understanding the rheological characteristics of the lunar regolith slurry is crucial to support the extrusion and deposition process during 3D printing. Momi et al. (2021b) investigated the rheological properties of mixtures composed of the JSC-1A simulant, water, NaOH (an alkaline agent), and urea (a superplasticizer). The result shows that these mixtures exhibit viscoplastic behavior, which can be accurately modeled using the Bingham rheological model (Bingham 1917).

According to the Bingham model, a material remains solid and does not flow until the applied shear stress  $\tau$ , exceeds a critical yield stress  $\tau_0$ . Once this threshold is surpassed, the material flows linearly with the shear rate. The relationship between viscosity  $\mu$ , and shear rate  $\dot{\gamma}$ , is described by the following equation:

$$\mu(\dot{\gamma}) = \begin{cases} \infty, & \text{when } \tau < \tau_0; \\ \eta_P + \frac{\tau_0}{\dot{\gamma}}, & \text{when } \tau \geq \tau_0; \end{cases} \quad (1)$$

where  $\eta_P$  is the plastic viscosity of the material. The viscosity becomes infinite when  $\tau < \tau_0$  or at zero shear rate, which typically causes numerical instabilities in the computational model. To overcome this, the following bi-viscous regularization of the Bingham material model is used (O'Donovan and Tanner 1984):

$$\mu(\dot{\gamma}) = \begin{cases} \mu_{M.app}, & \text{when } \dot{\gamma} < \dot{\gamma}_c; \\ \eta_P + \frac{\tau_0}{\dot{\gamma}}, & \text{when } \dot{\gamma} \geq \dot{\gamma}_c; \end{cases} \quad (2)$$

where  $\mu_{M.app} = \eta_P + \tau_0/\dot{\gamma}_c$  is the maximum apparent viscosity and  $\dot{\gamma}_c$  is the critical shear rate. The choice of  $\dot{\gamma}_c$  is arbitrary, yet it should be sufficiently small to lead to a relatively higher maximum apparent viscosity to restrict flow in regions where stress is below the yield stress, yet sizable enough to prevent prolonged computations and potential divergence problems. In this study,  $\dot{\gamma}_c = 0.1 \text{ s}^{-1}$  in accordance with Mollah et al. (2021).

The details of the regolith simulant-water slurry considered for this study and its rheological properties are summarized in Table 1.

**Table 1** Physical and rheological properties of the regolith simulant-water slurry; extracted from Momi et al. (2021b)

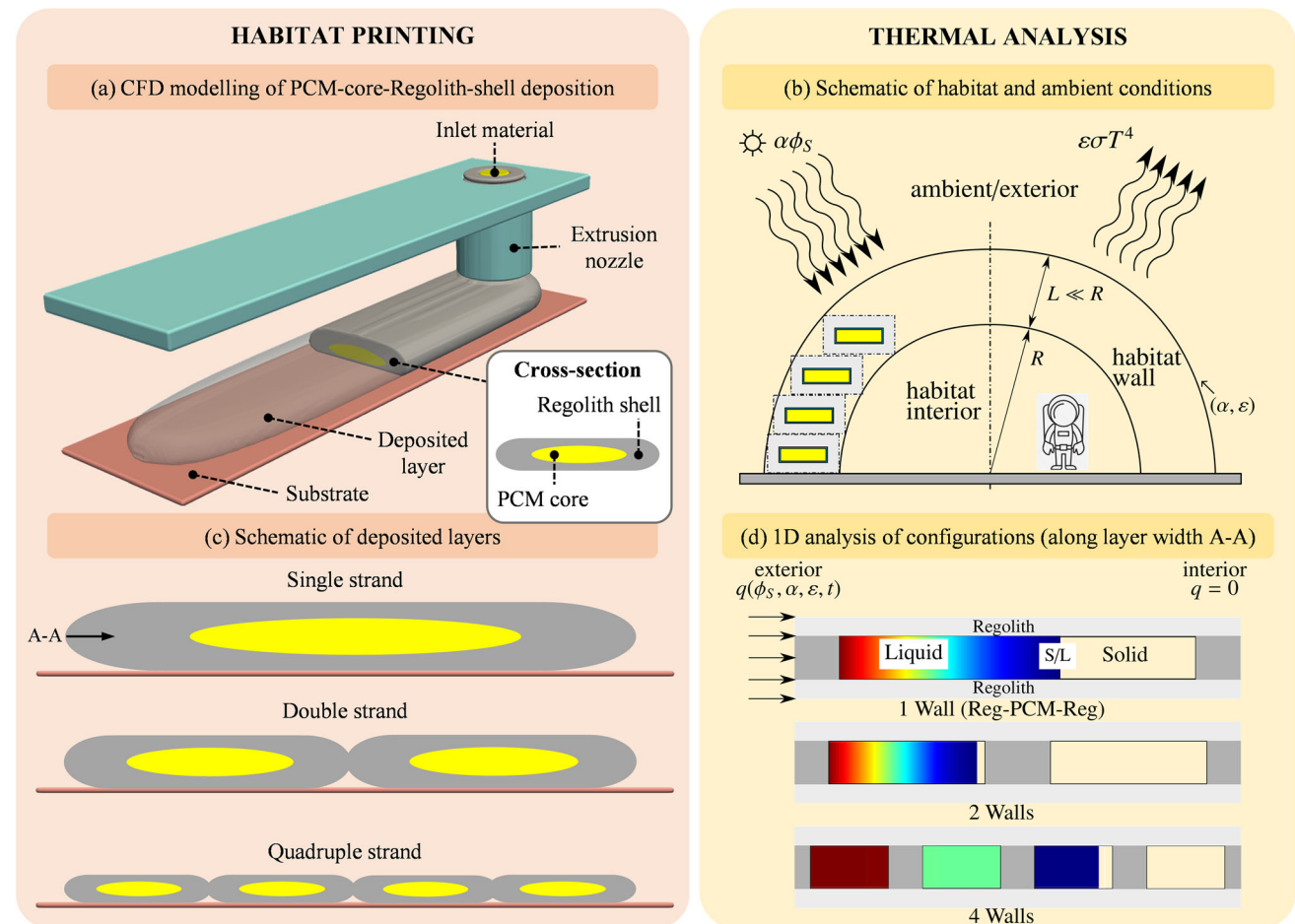
| Property                             | Value |
|--------------------------------------|-------|
| Regolith content, $w_R$ (wt%)        | 77.3  |
| Volume fraction (%)                  | 53.7  |
| Density, $\rho$ (kg/m <sup>3</sup> ) | 2042  |
| Dynamic yield stress, $\tau_0$ (Pa)  | 41    |
| Plastic viscosity, $\eta_P$ (Pa.s)   | 4.4   |

### CFD Modeling of PCM-core-Regolith-shell Deposition

The CFD model simulates the extrusion-deposition process of materials, PCM (in the core) and regolith (in the shell), under lunar gravity. The model includes a cylindrical extrusion nozzle with a diameter  $D$ , positioned at a distance  $h$  above a substrate (i.e., the nominal height of a layer), as illus-

trated in Fig. 1(a). The nozzle consists of two mass sources (in the core and the shell) that introduce material at volumetric flow rates  $Q_{\text{core}}$  and  $Q_{\text{shell}}$ , respectively, corresponding to an extrusion speed  $U$ . During deposition, the nozzle extrudes the PCM and regolith while moving at a constant printing speed of  $V$ . Table 2 summarizes the printing conditions in different simulation cases.

The size of the computational domain varies depending on the case considered. The domain has a length of  $9D$  and a height of  $(3D + h)$ , while the width differs across cases:  $4D$  for Case 2,  $9D$  for Case 3, and  $2D$  for all other cases. For detailed descriptions of the simulation cases, refer to Table 2. Wall boundary conditions, which prevent fluid flow, are applied to the top and bottom planes, while continuity boundary conditions are imposed on all other planes. The interface between the solid objects and the material is assigned to a no-slip boundary condition.



**Fig. 1** (a, b) Illustrations of (a) coaxial 3D printing, where layers consist of a regolith shell and a PCM core, and (b) a 3D-printed space habitat along with its ambient conditions. (c, d) Schematics depict-

ing (c) the deposited layers in ‘single’, ‘dual’ and ‘quadruple’ strand alternatives, and (d) the corresponding 1-dimensional domains used for thermal analysis



**Table 2** 3D printing cases: processing conditions and hydrostatic pressure

| Case ID | Nozzle diameter<br>$D$ (mm) | Layer height<br>$h$ (mm) | Geometrical ratio<br>$h/D$ (-) | Extrusion speed<br>$U$ (mm/s) | Volumetric flow rate<br>$Q_{\text{core}}$ (m <sup>3</sup> /s) | Volumetric flow rate<br>$Q_{\text{shell}}$ (m <sup>3</sup> /s) | Printing speed<br>$V$ (mm/s) | Speed ratio<br>$V/U$ (-) | Strand-to-strand distance<br>$S_{\text{toS}}$ (mm) | Hydrostatic pressure<br>$P_H$ (Pa) |
|---------|-----------------------------|--------------------------|--------------------------------|-------------------------------|---------------------------------------------------------------|----------------------------------------------------------------|------------------------------|--------------------------|----------------------------------------------------|------------------------------------|
| 1       | 50                          | 25                       | 0.5                            | 30                            | $1.47 \times 10^{-5}$                                         | $4.42 \times 10^{-5}$                                          | 30                           | 1                        | -                                                  | 81.97                              |
| 2       | 25                          | 12.5                     | 0.5                            | 30                            | $3.68 \times 10^{-6}$                                         | $1.10 \times 10^{-5}$                                          | 30                           | 1                        | $1.88 \times D$                                    | 40.99                              |
| 3       | 12.5                        | 6.25                     | 0.5                            | 30                            | $9.95 \times 10^{-7}$                                         | $2.69 \times 10^{-6}$                                          | 30                           | 1                        | $1.88 \times D$                                    | 20.49                              |
| 4       | 50                          | 25                       | 0.5                            | 20                            | $9.82 \times 10^{-6}$                                         | $2.95 \times 10^{-5}$                                          | 30                           | 1.5                      | -                                                  | 81.97                              |
| 5       | 50                          | 25                       | 0.5                            | 40                            | $1.96 \times 10^{-5}$                                         | $5.89 \times 10^{-5}$                                          | 30                           | 0.75                     | -                                                  | 81.97                              |
| 6       | 50                          | 20                       | 0.4                            | 30                            | $1.47 \times 10^{-5}$                                         | $4.42 \times 10^{-5}$                                          | 30                           | 1                        | -                                                  | 81.97                              |
| 7       | 50                          | 30                       | 0.6                            | 30                            | $1.47 \times 10^{-5}$                                         | $4.42 \times 10^{-5}$                                          | 30                           | 1                        | -                                                  | 81.97                              |

The flow of material is assumed to be transient, isothermal, and incompressible with a constant density  $\rho$ , so that it is governed by the following mass and momentum conservation equations:

$$\nabla \cdot \mathbf{u} = 0, \quad (3)$$

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (4)$$

where  $\mathbf{u}$  and  $p$  are the velocity and pressure fields,  $\boldsymbol{\sigma}$  is the deviatoric stress tensor, and  $\mathbf{g}$  is the gravitational acceleration vector with components  $(0, 0, -g)$ . Here,  $g = 1.62 \text{ m/s}^2$  is the gravity acceleration at the Moon's surface.

The deviatoric stress tensor is approximated with the following generalized Newtonian constitutive law:

$$\boldsymbol{\sigma} = 2\mu(\dot{\gamma}) \mathbf{D}_T, \quad (5)$$

where the apparent viscosity Eq. 2 described above is used. Here,  $\dot{\gamma} = \sqrt{2 \text{tr}(\mathbf{D}_T^2)}$  represents the strain rate magnitude and  $\mathbf{D}_T$  represents the deformation rate tensor, defined as:

$$\mathbf{D}_T = \frac{1}{2} \left( \nabla \mathbf{u} + (\nabla \mathbf{u})^T \right), \quad (6)$$

where the superscript  $T$  denotes the transpose.

The governing Eqs. 3 and 4 are discretized using the Finite Volume Method and solved with the commercial CFD software FLOW-3D<sup>®</sup> (V22R1; Flow Science Inc.). The solution technique of the developed model follows previous studies (Mollah et al. 2023b, 2025) except for the implementation of the two mass sources in the extrusion nozzle, allowing to simulate the simultaneous extrusion of two different materials (Regolith and PCM) in a shell-core manner. The model utilizes the FAVOR method (Fractional Area / Volume Obstacle Representation) to embed solid objects such as the nozzle, mass sources, and substrate within the computational domain. This approach integrates solid-geometry

effects into the governing equations, enabling accurate simulation of material extrusion through the moving nozzle and deposition onto the substrate. Additionally, the Immersed Boundary Method was employed to enhance the prediction of pressure and forces near solid objects. Momentum advection was calculated using an explicit upwind-difference technique, ensuring second-order accuracy in space and first-order accuracy in time. The pressure-velocity coupling in the governing equations was resolved using the GMRES (Generalized Minimum Residual) solver (Ashby et al. 1990; Barrett et al. 1994; Saad 1996), while the viscous stress terms in the momentum equation were handled through an implicit scheme. Free surface tracking in open channels was performed using the Volume-of-Fluid method (Hirt and Nichols 1981; Comminal et al. 2015). To maintain numerical stability, the time step size was dynamically adjusted based on the computed stability limit. However, an initial minimum time step of  $1 \times 10^{-5} \text{ s}$  was imposed to prevent numerical instabilities.

For single-material extrusion, this numerical model has been extensively validated in Spangenberg et al. (2021); Comminal et al. (2020) and Mollah et al. (2023a), where it accurately predicted the overall morphology of strands printed with cementitious materials, providing strong confidence in its ability to accurately describe material deposition. For coaxial extrusion, this is the first time the model is applied to simulate it and is yet to be validated experimentally; this validation will be undertaken elsewhere.

## Thermal Analysis

### Problem Statement and Hypotheses

The habitat concept is a semi-spherical (igloo-shaped) geometry of radius  $R$  that is built via 3D printing of multiple layers, stacked one-on-top-of-the-other, of a PCM core contained within a regolith shell; see Fig. 1(b). The use of PCM is considered for passive thermal control, so that the interior

temperature of the habitat is maintained within an adequate range for human life. Although it is not part of the current analysis, it is worth noting that the habitat interior will be pressurized for suitable habitable conditions.

Following the work of Borshchak Kachalov et al. (2023), a thermal analysis is performed on the cross-section of one of the printed layers of thickness  $L$  ( $\ll R$ ); see Fig. 1(c). Additionally, the reduced gravity on the lunar surface allows to use a simplified 1-dimensional model that neglects convection in the liquid PCM; see Fig. 1(d).

The habitat is located on the Moon, without any specific placement. For this reason, the thermal analysis is performed using generic characteristics of the lunar environment:

- The solar radiation flux  $\phi_S$  is the only heat source considered, modeled using the Moon's solar constant (Jet Propulsion Laboratory 2024):

$$\phi_S^{\max} = 1368 \text{ W/m}^2.$$

- At the external habitat wall, radiative heat exchange is governed by the thermo-optical properties — solar absorptivity  $\alpha$ , and infra-red emissivity  $\varepsilon$  — of the applied coating or paint, with values ranging between [0, 1].
- The period of the illumination cycle is  $\mathcal{T} = 28$  days, i.e. the mean solar day.
- Due to lunar rotational dynamics, the external boundary of the habitat undergoes cyclic illumination and eclipse periods of duration  $\mathcal{T}_i$  and  $\mathcal{T}_e$ , which are made dimensionless using  $\mathcal{T}$ :

$$\tau_i + \tau_e = 1.$$

- Although the eclipse fraction  $\tau_e$  depends on location and geography, unless otherwise stated, values of  $\tau_e = \tau_i = 0.5$  are used, corresponding to the (averaged) lunar eclipse duration.
- Heat exchange between the habitat and its surroundings occurs at its external boundary,  $x = 0$ . Since the habitat rotates with the Moon, the average solar flux received

over one illumination cycle can be approximated by:

$$\bar{\phi}_S = \frac{2}{\pi} \phi_S^{\max}.$$

Additionally, the following simplifications are considered in the model:

- The temperature at the inner wall boundary,  $x = L$ , denoted as  $T_i$ , is assumed to match the habitat's internal temperature, making this boundary adiabatic.
- The interface between regolith and PCM is assumed to be perfectly conducting.

Including the PCM requires the habitat to operate around the phase change temperature so as to exploit its latent heat. Currently, a wide variety of PCM is available on the market. Inorganic PCM, typified by hydrated salts and metals, exhibit high energy density and thermal conductivity, but can be corrosive, fact that limits their reliability. Besides, hydrate salts often suffer from subcooling. Organic PCM, typified by alkanes and fatty acids, have relatively low thermal conductivity, but they offer good chemical stability and do not require nucleating agents (Tatsidjodoung et al. 2013). Recently, Katish et al. (2024) analyzed the thermal stability of organic PCM under 10.000 thermal cycles, demonstrating stable thermo-physical properties and minimal mass loss below 150 °C. Considering these characteristics, together with a moderate melting temperature of 18 °C compatible with life conditions, the material n-hexadecane is selected from the family of organic PCM. Its thermo-physical properties are outlined in Table 3.

Additionally, the properties of raw and sintered 3D printed regolith are also included in this table (Fateri et al. 2019), with those of raw regolith used for simulations. Remark that, we envisage the geopolimerization of regolith, curing the material at ambient temperature (Zhan et al. 2025) or near the melting point of the binder (Bao et al. 2024). Therefore, this temperature typically remains below 100–200 °C, which is lower than 286 °C, the boiling point of n-hexadecane (Lide

**Table 3** Thermo-physical properties of raw and sintered regolith in solid state, and n-hexadecane; reproduced from Salgado Sanchez et al. (2020); Velez et al. (2015); Schreiner et al. (2016); Fateri et al. (2019)

| Property                                                                 | Regolith (raw) | Regolith (sintered) | nC16        |
|--------------------------------------------------------------------------|----------------|---------------------|-------------|
| Melting temperature, $T_M$ (°C)                                          | 1773           | –                   | 18          |
| Latent heat, $c_L$ (kJ kg <sup>-1</sup> )                                | 450            | –                   | 237         |
| Density, $\rho_{l/s}$ (kg m <sup>-3</sup> )                              | 1500           | 1570 - 2570         | 765/765     |
| Specific heat capacity, $c_{pl/s}$ (J kg <sup>-1</sup> K <sup>-1</sup> ) | 700            | 740                 | 2220/1950   |
| Thermal conductivity, $k_{l/s}$ (W m <sup>-1</sup> K <sup>-1</sup> )     | 0.0015         | 0.55 - 1.2          | 0.145/0.330 |

The subscripts ' $l$ ', ' $s$ ' refer to the liquid and solid phases, respectively. The properties of sintered regolith are presented as ranges, as the samples were manufactured using different techniques and are provided here for reference purposes only

2014). In these circumstances, the chemical stability of the PCM is assured over multiple melting-solidification cycles (Katish et al. 2024).

Another relevant consideration, arising from the absence of atmosphere and exposure to a low-pressure environment (approximately vacuum), is the associated variation in material properties, which can be challenging from a design perspective. Note, however, that the PCM remains embedded within the regolith shell and is not directly exposed to vacuum. Therefore, although some variation in PCM properties may occur, its impact is expected to be minimal. In contrast, the porosity of the shell may have a significant effect, which will be addressed in a separate study.

### Governing Equations

The dynamics of the habitat wall are described by the conservation of energy, which accounts for temperature variations (sensible heat) and the phase change (latent heat) in the PCM:

$$\rho c_p \frac{\partial T}{\partial t} + \rho c_L \frac{\partial f}{\partial t} = \frac{\partial}{\partial x} \left( k \frac{\partial T}{\partial x} \right), \quad (7)$$

where  $T$  is the temperature field,  $k$  and  $c_p$  are the thermal conductivity and heat capacity at constant pressure, and  $c_L$  is the latent heat. Recall that, for the regolith shell, only the contribution of the sensible heat is retained.

The local liquid fraction  $f$  is modeled as a step function depending on the  $T$  and smoothed over a temperature interval  $\delta_T$  around the solid-liquid phase change temperature  $T_M$ . This interval represents the *ushy region* where the solid ( $f = 0$ ) and liquid ( $f = 1$ ) phases coexist (Egolf and Manz 1994). According to previous works, a value  $\delta_T = 1$  K is selected (Salgado Sánchez et al. 2020; Garcia-Acosta et al. 2022).

The PCM is defined as a single phase, with its thermo-physical properties depending on  $T$  via the local value of  $f$ :

$$(\rho, c_p, k) = (\rho_s, c_{ps}, k_s)(1 - f) + (\rho_l, c_{pl}, k_l)f, \quad (8)$$

where subscripts ' $s$ ' and ' $l$ ' refer to solid or liquid, respectively.

At the external boundary of the habitat,  $x = 0$ , the heat exchange via radiation reads:

$$-k \frac{\partial T}{\partial x} = \underbrace{\alpha \bar{\phi}_S g(t)}_{\text{absorbed}} - \underbrace{\varepsilon \sigma_{\text{SB}} T^4}_{\text{radiated}}, \quad (9)$$

where the first term represents the absorbed heat, which depends on  $\alpha$  and  $\bar{\phi}_S$ . The most-right term describes the radiated heat, modeled via the Stefan-Boltzmann law where  $\sigma_{\text{SB}} = 5.67051 \times 10^{-8} \text{ W/(m}^2 \text{ K}^4)$  is the Stefan-Boltzmann constant. In addition, the function  $g(t)$  is used to define the

cycles of illumination/eclipse. Here,  $g(t)$  is defined as a step function:

$$g(t) = \begin{cases} 1, & t \in [nT, nT + T_i); \\ 0, & t \in [nT + T_i, (n+1)T); \end{cases} \quad (10)$$

where  $n = 0, 1, 2, \dots$  refers to the  $(n+1)$ th solar cycle.

At the inner boundary of the habitat wall,  $x = L$ , an adiabatic condition is assumed:

$$\frac{\partial T}{\partial x} = 0, \quad (11)$$

consistent with the hypotheses described above.

Taking the wall thickness  $L$  and the timescale for thermal diffusion in the PCM  $\tau = L^2/\alpha$ , where  $\alpha = k/(\rho c_p)$ , as characteristic values, the thermal dynamics depends on the dimensionless solar period:

$$\hat{T} = T \left( \frac{\alpha}{L^2} \right). \quad (12)$$

Below, we use  $\hat{T}$  to discuss the effect of varying thermo-physical properties of the PCM.

### Numerical Model

The finite-element-based software COMSOL Multiphysics is used to solve the habitat thermal response. The initial condition for  $T$  is  $0^\circ \text{C}$  in the entire wall.

The habitat wall is meshed using a maximum element size of 0.75 mm for the regolith, and 0.25 mm for the PCM, to better capture the solid-liquid front movement within the PCM domain. The time evolution is integrated using a second-order Backward Differentiation Formulae scheme, with a maximum time step of 500 s (i.e., approximately 5000 steps per cycle  $T$ ) and a relative tolerance of  $10^{-4}$ . Recall that these values are selected based on dedicated convergence analyses using the error in the liquid fraction of PCM ( $\mathcal{L}$ ):

$$E = \frac{1}{N_C T} \int_0^{N_C T} |\mathcal{L} - \mathcal{L}^*| dt, \quad (13)$$

as metric, where  $N_C$  is number of solar cycles simulated and  $\mathcal{L}^*$  refers to the finest case simulated.

For further details on the thermal modeling and convergence tests, the reader is referred to Borshchak Kachalov et al. (2023).

### 3D Printing Results

In this section, the effect of printing parameters on the shell-core structure is analyzed via 3D CFD simulations, as

described in Section “[Methodology](#)”, with cross-sectional shapes and the internal regolith-PCM structure of deposited layers as a simplified 1D domain for thermal analysis. As shown in Table 2, seven different cases were examined, each involving variations in printing conditions (i.e., the diameter of the nozzle  $D$ , layer height  $h$ , and extrusion speed  $U$ ).

As established in prior studies, printing conditions significantly influence the morphology of the deposited layer (Mollah et al. 2021; Šeta et al. 2023). Two key non-dimensional parameters can be defined: the geometrical ratio  $h/D$ , which is the ratio of the gap to the nozzle diameter, and the speed ratio  $V/U$ , which is the ratio of nozzle speed to extrusion speed. Typically, increasing the geometrical ratio, either by using a smaller nozzle or a larger layer height, results in more rounded strands with a smaller width (i.e., a thinner habitat wall in this context). Similarly, a more rounded strand shape can be achieved by increasing the speed ratio, either by increasing the nozzle movement speed or by reducing the extrusion speed (i.e., the volumetric flow rate of the material). However, it is important to clarify that changing the speed ratio inherently alters the amount of material in the cross-section of the strand. In contrast, modifying the geometrical ratio by adjusting the layer height does not necessarily have the same effect. These parameters varied systematically using the CFD simulations.

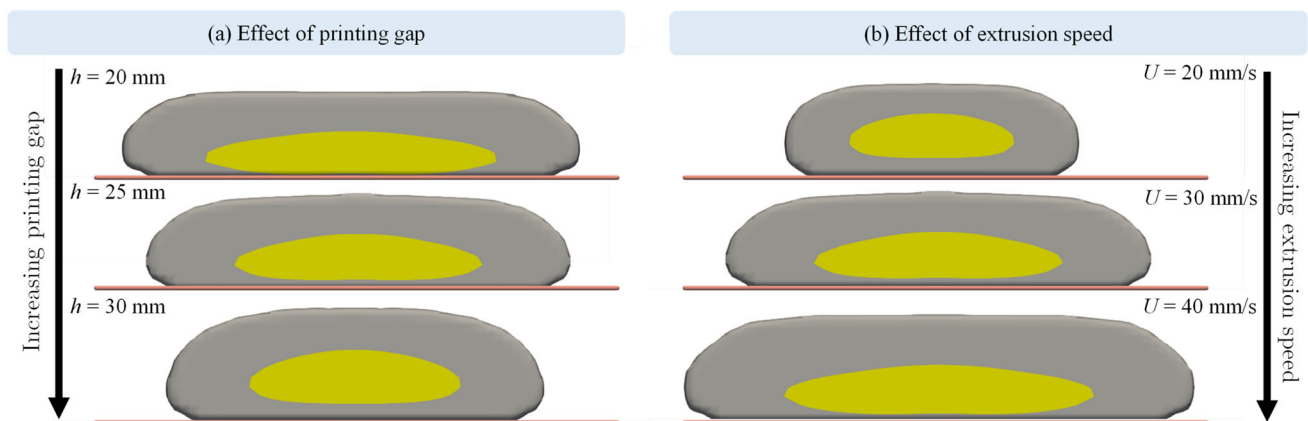
Since the extrusion system employs a coaxial nozzle, changes in printing conditions not only affect the strand shape but also alter its internal structure, specifically the position of the PCM core inside regolith shell. Anticipate that the width of this PCM core may significantly impact the thermal response of the regolith-PCM wall, emphasizing the importance of precise control over printing parameters and resulting strand shape/internal structure.

### Single Regolith-PCM Strand: Effect of Printing Parameters

In 3D printing, the geometrical and speed ratios are typically set to 0.5 and 1, respectively, to ensure adequate material deposition while maintaining a shape suitable for subsequent layers. Based on this convention, we established Case 1; see Table 3. A 50 mm nozzle diameter is initially selected as a reasonable starting point, a value commonly used in large-scale concrete 3D printing. This choice resulted in single-pass walls approximately 100 mm wide (exactly 96.2 mm). In the subsequent thermal analysis, this width will serve as the baseline thickness for performance evaluation across multiple cases.

Given that the coaxial nozzle is designed such that the core diameter is half the total diameter,  $D_{\text{core}} = D/2$ , the expected material distribution along the strand width is 25%-50%-25% (reg%-PCM%-reg%). Indeed, the simulated distribution closely follows this expectation, with a 49 mm wide PCM core, symmetrically surrounded by the regolith shell of 23.6 mm width on each side, resulting in 24.5%-51%-24.5% wall distribution. Obviously, this distribution is not uniform across the 2D cross-section. However, to ensure consistency across different cases, we identify a reference line at approximately two-thirds of the PCM height. This line is later used in Section “[Thermal Analysis](#)” to represent the wall composition. As can be seen in Fig. 2, the PCM core is positioned in the bottom-central region of the strand (in yellow). The downward shift of the PCM is attributed to the flow behavior of the regolith beneath the PCM core, which spreads laterally at the substrate, making space for the PCM core to be displaced.

In Fig. 2(a), only the layer height is varied, while all other conditions remain unchanged. This height was adjusted



**Fig. 2** Strand morphology and internal shell-core distribution of regolith-PCM (gray–yellow regions) with different (a) gap height  $h$ , (b) extrusion speed  $U$



between  $h = 20$  mm, 25 mm (baseline case), and 30 mm, corresponding to geometrical ratios of  $h/D = 0.4, 0.5$ , and  $0.6$ , respectively. As previously stated, changing only the layer height does not affect the total amount of deposited material. However, as observed in the figure, when  $h = 20$  mm, the strand becomes significantly wider and the PCM core expands, resulting in a distribution of 21.5% - 57% - 21.5%. Conversely, for  $h = 30$  mm, the strand is narrower, and the material distribution remains similar to that of the baseline  $h = 25$  mm case.

Figure 2(b) shows that when changing extrusion speed from  $U = 20$  mm/s to  $U = 40$  mm/s, corresponding to the speed ratios of  $V/U = 1.5$  to  $0.75$ , the strand morphology and internal structure are strongly affected. In this case, the amount of deposited material changes accordingly. As expected, increasing the extrusion speed results in a wider strand. Since the layer height remains the same, the excess material is pushed laterally. This effect is evident at  $U = 40$  mm/s, where the strand width is significantly greater than in the other two cases, leading to changes in the internal structure. Specifically, the PCM core accounts for 55% of the strand width in this case, an absolute 4% increase with respect to the baseline case; see Table 4. Overall, it can be concluded that when  $V/U < 1$  and  $h/D > 0.5$ , the distribution in the wall deviates more from the one that is found initially in the nozzle.

An interesting comparison can be made between the two cases with the largest strand width:  $h = 20$  mm and  $U = 40$  mm/s. As seen in Fig. 2, the regolith shell on top of the PCM is thinner in the  $h = 20$  mm case than in the  $U = 40$  mm/s case, due to the lower layer height. While this difference does not significantly affect the 1D thermal analysis (as the regolith-PCM distribution in a single line is considered to be similar), it could become relevant when considering 2D or 3D effects in future analyses.

### Multi-strand Wall Configuration

3D printing typically involves a layer-by-layer deposition, both side-by-side and on-top-of-each-other. With this in

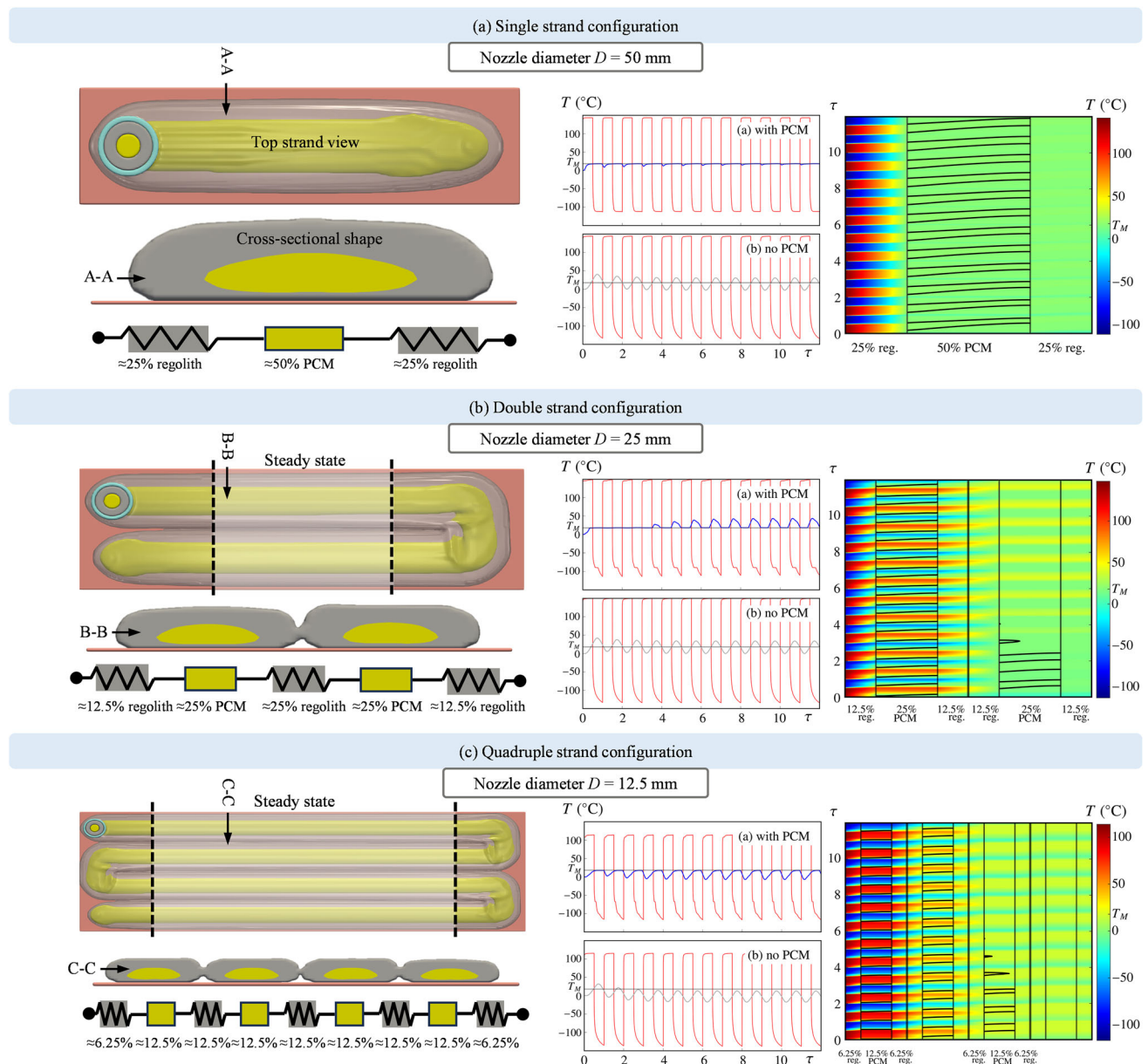
mind, we analyzed different strand configurations to achieve varying internal wall structures of the habitat, serving again as a reference for the subsequent thermal analysis presented in Section “Thermal Analysis”. This approach allowed to generate single, double, and quadruple sandwich structures (strands) of regolith-PCM, as illustrated in Fig. 1(c) and the left panels of Fig. 3.

To ensure that the double and quadruple strand configurations maintained a similar total width and could be compared to the single-strand configuration specially during the 1D thermal analysis, the nozzle diameter was reduced. Since the geometrical ratio was kept constant at  $h/D = 0.5$ , the layer height was also reduced accordingly. Specifically, to achieve the desired width in the double strand configuration, the nozzle diameter was reduced to  $D = 25$  mm, whereas it is  $D = 12.5$  mm for the quadruple strand configuration. For both configurations, the strand-to-strand distance was set to  $1.88 D$  before printing. This spacing was chosen to ensure proper strand-to-strand contact after deposition while preventing excessive squeezing and penetration between them.

The deposited strands resulted in a total width reduction of 2.9 mm in the quadruple strand configuration and 2.5 mm in the double strand configuration. This reduction is likely due to higher hydrostatic pressure in the case with larger nozzle diameters as the layer heights increase, which promotes lateral material flow. As seen in Table 2, the hydrostatic pressure in a single strand configuration is twice that of the double strand configuration and four times greater than in the quadruple strand configuration. This becomes even more significant when considering that the hydrostatic pressure in the deposited strands exceeds the material’s yield stress of 41 Pa, see Table 1 for details. Although this effect is not explicitly analyzed here, it suggests that the material would likely continue to flow laterally until reaching an equilibrium between yield stress and hydrostatic pressure, as described by Mollah et al. (2025). Thus, while lower hydrostatic pressure in smaller nozzles improves structural stability, it also slows down the printing process. A potential downside of the multi-wall configuration is that it requires bead interfaces, creating small voids between adjacent walls. These

**Table 4** Summary of total thickness, PCM thickness, and composition under different printing conditions

| Condition     | Wall thickness (mm) | PCM thickness (mm) | Approx. composition (%) |
|---------------|---------------------|--------------------|-------------------------|
| $U = 20$ mm/s | 67.0                | 34.4               | 24.5–51–24.5            |
| $U = 30$ mm/s | 96.2                | 49.8               | 24.5–51–24.5            |
| $U = 40$ mm/s | 116.4               | 64.0               | 22.5–55–22.5            |
| $h = 20$ mm   | 103.6               | 59.2               | 21.5–57–21.5            |
| $h = 25$ mm   | 96.2                | 49.8               | 24.5–51–24.5            |
| $h = 30$ mm   | 83.6                | 42.8               | 24.5–51–24.5            |



**Fig. 3** Results for the (a) single, (b) double and (c) quadruple strand configurations: (left) 3D printed cross-sectional shapes, (right) thermal response. In right panels, contours showing the time evolution of the temperature field along the wall and associated temperature evolutions

at the external (red) and internal boundaries with (gray) or without (blue) PCM are illustrated for  $\varepsilon = 0.2$ ,  $L = 100$  mm,  $\tau_e = 0.5$ , and: (a) single strand,  $(\alpha/\varepsilon) \simeq 2$ ; (b) double strand,  $(\alpha/\varepsilon) \simeq 2.1$ , and (c) quadruple strand,  $(\alpha/\varepsilon) \simeq 1.5$

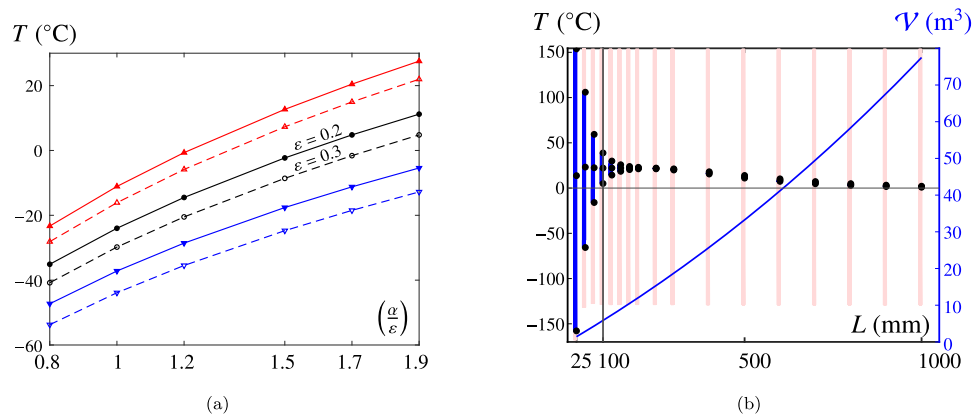
voids may interfere with thermal performance, further highlighting the trade-offs between different printing strategies.

## Thermal Analysis

We analyze now the thermal response of the habitat, under the assumptions and hypotheses described in Section “[Methodology](#)”. Results are first presented for a wall made of just regolith, which is likely the most evident option to built a

lunar habitat, describing the effect of different design parameters. This reference case is then used to support the analysis of integrating n-hexadecane as PCM, and the supplemental effect of considering multiple regolith-PCM layers to construct the habitat.

First, we find it important to remark that, after several illumination/eclipse cycles, the temperature evolution of the wall reaches a steady, periodic state (Borshchak Kachalov et al. 2023), as illustrated in the right panels of Fig. 3, where temperature evolutions at the external (red) and internal



**Fig. 4** (a) Dependence on  $(\alpha/\varepsilon)$  of the maximum (red), mean (black) and minimum (blue) steady temperatures at the internal wall boundary. Results are shown for  $\varepsilon = 0.2$  (solid lines) and  $0.3$  (dashed lines), with  $L = 100$  mm and  $\tau_e = 0.5$ . (b) Effect of  $L$  on the steady temperature fluctuations at the internal boundary (left axis) and the required regolith

boundaries of the habitat are shown for three representative cases with (blue) and without PCM (gray curves); these results will be further discussed below. At any wall depth, therefore, the periodic evolution of  $T$  can be characterized by their maximum, minimum, and mean values. For the main purpose of controlling the internal habitat temperature  $T_i$  within a suitable range for living conditions, we use the representative values  $T_i^{\max}$ ,  $T_i^{\min}$ , and  $T_i^{\text{mean}}$  to characterize the dynamics.

Hereafter, the effect of the main design parameters are discussed. We focus first on the influence of the thermo-optical properties  $(\varepsilon, \alpha)$  at the external wall boundary, and the thickness of the habitat wall,  $L$ . Note that these parameters can be selected regardless the habitat's location and environment on the lunar surface.

### Effect of Thermo-optical Properties and Wall Thickness

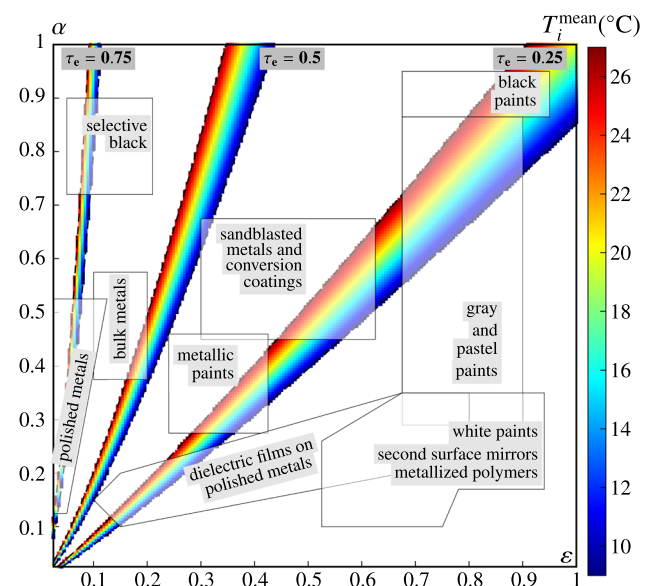
In Fig. 4(a), the maximum, minimum and mean values of  $T_i$  are illustrated for varying  $(\varepsilon, \alpha)$  and a wall thickness  $L = 100$  mm; see Section “3D Printing Results” for details on this selection. By reducing (increasing) the emissivity, as done in the curves for  $\varepsilon = 0.2$  (solid),  $0.3$  (dashed), the internal habitat temperature increases (decreases), since  $\varepsilon$  controls the relative amount of radiated heat at the external habitat boundary. On the other hand, varying  $(\alpha/\varepsilon)$  at constant  $\varepsilon$  allows to control the relative quantity of absorbed solar radiation so that the internal habitat temperature increases or decreases as  $\alpha$  does.

Considering these results, one can select a *bulk metal* coating for the external habitat boundary. This coating is characterized by  $(\varepsilon, \alpha) = (0.17, 0.37)$  (see Fig. 5), achiev-

ing a mean temperature  $T_i^{\text{mean}} \approx 22$  °C. Note, however, that the selection of  $(\varepsilon, \alpha)$  has little effect on the observed fluctuations,  $(T_i^{\max} - T_i^{\min})$ . Indeed, the amplitudes shown in Fig. 4(a) are larger than 20–30 °C, characterizing a temperature envelope that may not be compatible with human life conditions.

The selection of  $(\varepsilon, \alpha) = (0.17, 0.37)$  is now used to analyze the effect of the wall thickness  $L$ , which plays a pivotal

role in the design of the habitat.



**Fig. 5** Contours of the mean internal temperature as a function of  $(\varepsilon, \alpha)$ . The colored region represents values within the range [9, 27] °C. Results are shown for three eclipse fractions:  $\tau_e = 25, 50$  and  $75\%$ . Additionally, regions corresponding to thermo-optical properties of conventional paintings and coatings are overlaid, adapted from Tachikawa et al. (2022)

role in dampening these thermal fluctuations. In Fig. 4(b), the effect of  $L$  is illustrated along with the required volume of regolith  $\mathcal{V}$  to build a semi-spherical habitat of inner radius 3 m (Ongil et al. 2024). The ranges of temperature experienced at the exterior (red) and interior boundaries (blue) are indicated by vertical lines; circular markers are used to highlight the maximum, minimum and mean values of  $T_i$ .

At the external boundary, temperature variations show no significant dependence on  $L$ . At the internal boundary, however, these variations are strongly influenced by wall thickness. For thin walls, approaching the limit  $L \rightarrow 0$ , the regolith volume within the wall is minimal, resulting in an almost negligible temperature difference between the internal and external boundaries — both evolving nearly identically over time. As  $L$  increases, temperature fluctuations are progressively dampened, while  $T_i^{\text{mean}}$  remains nearly unchanged, aligning with the mean temperature at the external boundary. For  $L > 200$  mm, fluctuations disappear entirely, maintaining the inner boundary at an almost constant temperature; this requires a volume of regolith  $\mathcal{V} > 12.1 \text{ m}^3$ , i.e., a regolith mass of roughly 20 tons, on the order of the typical weight of small houses. For  $L > 300$  mm ( $\mathcal{V} > 18.7 \text{ m}^3$ ), the thermal insulation achieved by the regolith is such that  $T_i$  is almost insensitive to the external heat flux. Indeed, for even thicker walls,  $T_i$  remains at its initial condition (0 °C), confirming such insulation.

Although a complete thermal insulation may seem advantageous, it presents significant drawbacks:

- (i) The large amount of regolith required, along with the costs associated with its extraction and preparation.
- (ii) The insulation would necessitate additional thermal control strategies to manage the internal heat dissipation from crew activities and electronic equipment.
- (iii) The increased wall weight could impose substantial structural loads on the habitat base, posing challenges for 3D printing too.

Therefore, an optimal wall thickness in the range  $L \in [200, 300]$  mm is recommended, balancing costs and reduced temperature fluctuations. Besides this, and despite the quantity of regolith is not a constraint in terms of availability, it would influence significantly the time required to construct the habitat, as the material must be extracted from the lunar soil, mixed with the binder, printed at a speed that ensures the desired mechanical properties, and then cured. Ideally, this entire sequence should be carried out during daylight hours, allowing the use of solar panels to power the equipment and facilitate thermal post-treatment.

Considering a wall thickness within this range, so that  $T_i$  becomes insensitive to it, and the potential dependence on  $(\varepsilon, \alpha)$  shown in Fig. 3, one can construct contours of  $T_i^{\text{mean}}$  as function of  $(\varepsilon, \alpha)$ . In Fig. 5, these contours are

illustrated for the temperature interval  $T_i^{\text{mean}} \in [9, 27] \text{ }^\circ\text{C}$ , a representative range compatible with human life. The figure includes labeled regions of  $(\varepsilon, \alpha)$  that correspond to conventional paintings and coating materials used for passive thermal control (Tachikawa et al. 2022). Considering these materials, it is important to note that only labeled regions of  $(\varepsilon, \alpha)$  that intersect with the contours are feasible from a thermal control perspective, i.e., one can select a material available in the market to control the internal habitat temperature in a passive manner. In this sense, the reference case of a 50% eclipse fraction ( $\tau_e = 0.5$ ) requires the use of either *bulk metals* (unpolished) or *sandblasted metals and conversion coatings* to achieve a suitable mean temperature in the habitat interior. This region of parameters can be characterized by  $(\alpha/\varepsilon) \simeq 2.5$ .

Recall that these results correspond to raw regolith. Note, however, that they can vary after geopolymerization, which includes the addition of a binder, the extrusion process, and the thermal post-treatment; see Table 3. To anticipate this thermal response, one can use the dimensionless period  $\hat{T}$ , adapting the wall thickness  $L$  for varying thermo-physical properties so that  $\hat{T}$  and the associated mean internal temperature  $T_i^{\text{mean}}$  remain unchanged.

For completeness, simulations for  $\tau_e = 0.75, 0.25$  are used to construct associated contours of  $T_i^{\text{mean}}$ , included also in Fig. 5. For  $\tau_e = 0.75$ , the temperature range is shifted to smaller values of  $\varepsilon$  as the longer eclipse time requires less heat evacuation. In this case, *selective black* paintings and *polished metals*, characterized by  $(\alpha/\varepsilon) \simeq 10$ , have adequate thermo-optical properties and provide the desired thermal control. If the eclipse percentage is smaller, as with  $\tau_e = 0.25$ , a suitable habitat temperature can be obtained also for coatings with large emissivity values and a smaller thermo-optical relationship  $(\alpha/\varepsilon) \simeq 1.1$ .

To summarize, results indicate that an adequate thermal design that assures conditions for human life is feasible. First, one should select adequate paintings or coatings that balance the energy exchange at the external boundary during illumination and eclipse, thus controlling the mean habitat temperature. Then, one should select the minimum wall thickness able to reduce temperature fluctuations within an admissible range, so that the required quantity of regolith is minimized as much as possible.

These basic guidelines, however, require further assessment. On the one hand, the precise thermo-physical properties of regolith are unknown and would also depend on the manufacturing process and resulting characteristics of the building material (roughness, porosity, etc.). Other relevant issue is related to the natural degradation of the absorptivity, as it will affect the mean temperature in no small proportion considering the typical duration of lunar (or extraterrestrial) missions, on the order of years (Jaworske et al. 2008; Sharma and Sridhara 2012; Zinecker et al. 2023).



Based on these results, one can envision the integration of PCM within the regolith wall as a promising passive solution that improves the design and thermal performance of the habitat. The associated analysis is presented hereafter.

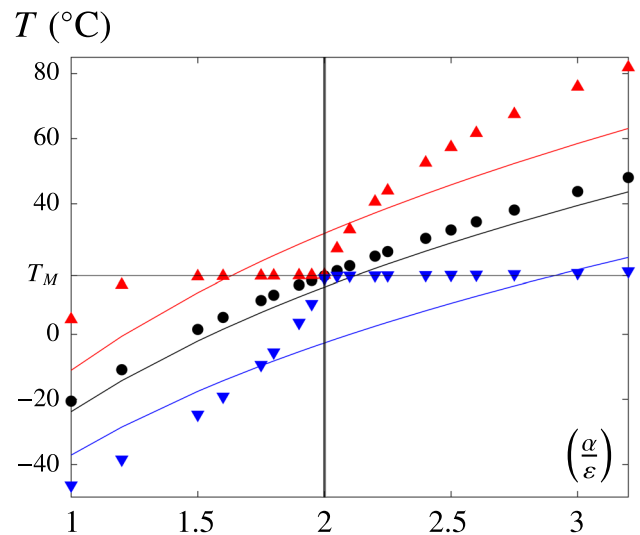
### Analysis of the Regolith-PCM wall

The nearly isothermal nature of melting and solidification makes the integration of PCM a promising option for passive thermal regulation, enhancing thermal inertia and balancing the energy budget of the habitat. A prior study explored the potential of n-alkanes for passive temperature control (Borshchak Kachalov et al. 2023). More recently, the same concept was experimentally tested using scaled laboratory experiments to analyze the effect of integrating a phase-changing material into an igloo-shaped habitat (Ongil et al. 2024). Both analyses suggest the ability of the PCM to stabilize the inner habitat temperature close to the melting point, with little to no fluctuations.

In this section, we investigate the thermal performance of a regolith wall integrating n-hexadecane, an organic PCM with a melting temperature  $T_M \simeq 18^\circ\text{C}$  compatible with living conditions; see Table 3. Note that the value of  $T_M$  will play a critical role since, for the PCM to be effective, it should experience repeated cycles of melting and solidification, maintaining the habitat temperature around  $T_M$ . For brevity, we refer to this type of habitat wall, made of a PCM core and a regolith shell, as reg-PCM-reg in the remainder of this manuscript. In this analysis, the PCM is placed symmetrically at the center of the wall and occupies a 50% of its thickness  $L$ ; see Section “3D Printing Results”. This setup corresponds to the ‘single strand’ case of panels (c, d) in Fig. 1 and panel (a) of Fig. 3.

Following the previous discussion, the effect of  $(\varepsilon, \alpha)$  is introduced first. In Fig. 6, the maximum (red), minimum (blue) and mean values (black markers) of  $T_i$  are illustrated for a wall thickness  $L = 100$  mm and  $\varepsilon = 0.2$ ; this permits a direct comparison with Fig. 4 and the reference case of just regolith (solid curves). Results indicate three different regimes:

- For  $(\alpha/\varepsilon) < 1.5$ , the balance between the absorbed and radiated heat is not enough to drive the phase change, and the PCM remains in solid state throughout repeated solar cycles. In this regime, since the solid PCM is more conductive than regolith, the effective thermal conductivity of the wall increases and, therefore, the mean temperature is slightly larger, displaying larger fluctuations as well.
- For  $(\alpha/\varepsilon) > 3$ , conversely, the energy balance at the external boundary saturates the PCM so that it works in liquid state throughout repeated solar cycles. Again, this regime is characterized by a larger effective conductivity of the wall compared to the case of just regolith and, there-



**Fig. 6** Dependence on  $(\alpha/\varepsilon)$  of the maximum (red), mean (black) and minimum (blue) steady temperatures at the internal regolith-PCM wall boundary. Results for just regolith are superimposed for the same parameters (solid lines), with the vertical line indicating the optimal value of  $(\alpha/\varepsilon)_{\text{opt}} = 2$

fore, the mean temperature is slightly larger, displaying also larger fluctuations.

- For  $1.5 < (\alpha/\varepsilon) < 3$ , the energy balance allows the PCM to be cycling between solid and liquid during a certain lapse of the solar cycle, so that it increases thermal inertia and reduces fluctuations compared to the case of just regolith. For  $(\alpha/\varepsilon) \simeq 2$ , in particular, the quantity of melted PCM during illumination equals that solidified during eclipse, stabilizing the interior temperature at  $T_M$ . This optimal value is similar to the estimate (Borshchak Kachalov et al. 2023):

$$\left(\frac{\alpha}{\varepsilon}\right)_{\text{opt}} = \frac{\sigma_{\text{SB}}}{\bar{\phi}_S} T_M^4 \tau_i^{-1}, \quad (14)$$

where a wall made of just PCM was analyzed. Here,  $\sigma_{\text{SB}}$  and  $\bar{\phi}_S$  are the Stefan-Boltzmann constant and average solar flux, as defined in Section “Methodology”.

Note that, for values close  $(\alpha/\varepsilon)_{\text{opt}}$ , there are some energy imbalances that cause the partial saturation of the PCM, either in solid or liquid, degrading PCM performance. Despite this, temperature fluctuations in this neighborhood exhibit amplitudes smaller than those observed with just regolith, still justifying the integration of the PCM. Besides, the presence of the PCM helps overcome the problem of absorptivity degradation in time, as the value of  $(\alpha/\varepsilon)$  has more room for variation, without compromising the achieved temperature control in a significant manner.

The right panels of Fig. 3(a) illustrate the time evolution of the regolith-PCM wall temperature for  $(\alpha/\varepsilon)_{\text{opt}}$ . The temper-



ature within the wall is shown as a contour, where the black line represents the location of the solid-liquid front, and the temperatures at the external (red) and internal (blue) boundaries are shown in the companion plot, where it is evident that  $T_i$  is almost constant and matches the value of  $T_M$ . For comparison, the temperature evolution at the wall boundaries for the case of just regolith are illustrated, where one can clearly observe that  $T_i$  (gray) presents a fluctuation  $\Delta T_i \sim \mathcal{O}(10)$  K.

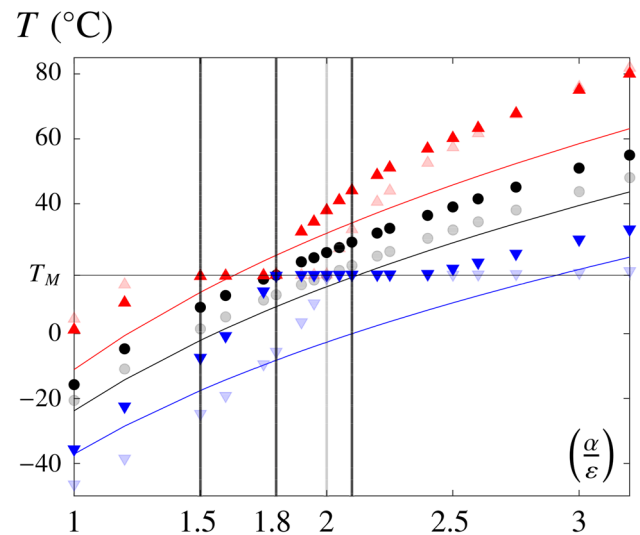
These results strongly support the integration of the PCM within the habitat wall to improve its passive thermal control. In line with this, note that the wall thickness can be reduced as soon as the quantity of PCM is enough to help balancing the energy exchange. Therefore, this either reduces the relative quantity of regolith required to construct the habitat, or allows for increasing the relative size/number of constructed habitats.

### Multiple Regolith-PCM Strands

We now analyze the possibility of using  $N$  strands of regolith-PCM to construct the wall. Sketches of these cases for  $N = 2, 4$ , labeled as ‘dual strand’ and ‘quadruple strand’, are illustrated in panels (c, d) of Fig. 1 and panels (b, c) of Fig. 3, respectively. The total wall thickness  $L$  is preserved, while the length of each strand is  $L' = L/N$ ; also, each strand inherits the previous distribution of 25-50-25% of regolith-PCM-regolith. Thus, the quantity of materials is also preserved with respect to the  $N = 1$  wall configuration for the same  $L$ . These multi-strand wall configurations are simulated assuming perfectly conducting contact between blocks, and approximated by placing the strands one after the other.

In Fig. 7, the steady internal temperatures (markers) are represented for the double strand configuration as a function of  $(\alpha/\varepsilon)$ . Results for just regolith (solid lines) and the  $N = 1$  case (reg-PCM-reg) analyzed above (light markers) are superimposed for visual comparison. Qualitatively, the dynamics for the double strand and  $N = 1$  cases are analogous, except for a slightly larger mean temperature and reduced fluctuations. The optimal selection of thermo-optical properties, however, is notably displaced to  $(\alpha/\varepsilon)_{N=2} = 1.8$ . For this case with  $\varepsilon = 0.2$ , this corresponds to a reduction of the absorptivity of  $\Delta\alpha = 0.04$ , which should be considered together with the available coating materials.

As discussed above when analyzing the effect of the PCM, the underlying reason for these changes is the variation in the effective thermal conductivity. Even though the quantity of regolith and PCM do not change, its distribution does, been now more homogeneous. For this reason, by increasing the number of strands  $N$ , the steady characteristic temperatures



**Fig. 7** Dependence on  $(\alpha/\varepsilon)$  of the maximum (red), mean (black) and minimum (blue) steady temperatures at the internal regolith-PCM wall boundary made of two blocks. Results for just regolith (solid lines) and one regolith-PCM block (light markers) are superimposed for comparison. Vertical lines correspond to  $(\alpha/\varepsilon) = 1.5, 1.8, 2.1$

will shows small and barely notable variations, as the total thermo-physical properties are already reasonably homogeneous; this is consistent with the results for  $N = 4$ , not represented in the figure for simplicity. Therefore, from a thermal control point of view, one can also consider the use of micro-encapsulated PCM, homogeneously distributed along the wall in the same proportion, scenario that would somewhat match the case of  $N \gg 1$ .

The evolution of the solid-liquid front within the PCM and the temperature field are represented for the double and quadruple strand configurations in Fig. 3. Now, these results are shown for values of  $(\alpha/\varepsilon)$  slightly different than the optimal to illustrate the partial saturation of the PCM discussed above; these are displaced to lower and larger values (marked by vertical lines in Fig. 7) for  $N = 2, 4$ , respectively.

First, from the contour, we observe that the PCM within the strand near the external boundary always undergo the phase change, so their latent heat is exploited in all cycles. On the other hand, and because the working point is not optimal, the remaining block temperatures present an upper or lower limit marked by the phase change temperature; this can be observed on the temperature evolution at the internal boundary as well. Note that in the first cycles, before the steady, periodic state is reached,  $T_i$  is maintained at the phase change temperature, which is the goal of this thermal control (the same behavior can be observed for  $N = 1$  near the optimal point). Once again, this supports the idea of overcoming

the degradation of  $\alpha$  by having initial conditions with slightly lower value.

Additionally, the use of several strands may allow the selection of different PCM (near the external and internal boundaries) in case it will be favorable to enhance the thermal control once heat sources are included within the habitat, i.e. crew or electronics. Finally, recall that this configuration requires smaller nozzle diameters for 3D printing, and wall thickness  $L$  is thus determined by the number  $N$  of printed strands.

## Conclusion

We presented here a coaxial 3D printing approach to integrate the deposition of regolith and PCM, forming a shell-core structure, with particular focus on lunar habitats. In this configuration, the regolith shell provides structural integrity, while the PCM core helps regulate the interior habitat temperature in a passive manner. The manuscript consisted of two main parts: CFD simulations of the printing process, incorporating both regolith and PCM, and the thermal analysis of the resulting 3D-printed strands.

3D printing simulations indicate that the (internal) regolith-PCM, shell-core structure of the strands can be tailored by modifying key printing parameters, such as adjusting the extrusion speed to change the amount of extruded material, or altering the layer height without affecting material deposition. Reducing the layer height from 25 mm to 20 mm increases the thickness of the PCM material by approximately 10 mm, while the total layer thickness increases by about 7 mm. This indicates that the PCM core expands relatively more than the regolith shell. A similar trend is observed when the extrusion speed is increased. Furthermore, the CFD model was used to simulate multi-strand deposition for different nozzle diameters and layer heights, resulting in double and quadruple strand configurations that match the initial total width of the single strand case. These configurations provided the internal morphology of the wall and a representative regolith-PCM distribution that facilitated the subsequent thermal analysis of the habitat.

The thermal analysis considered first a regolith-based habitat, evaluating its thermal performance for key design parameters: the infra-red emissivity  $\varepsilon$ , solar absorptivity  $\alpha$ , and wall thickness  $L$ . Lower  $\varepsilon$  reduces radiative heat evacuation, increasing internal temperatures, while variations in  $\alpha$  control absorbed solar radiation. A metal coating with  $(\varepsilon, \alpha) = (0.17, 0.37)$  was shown to achieve a mean internal temperature of 22°C, with significant temperature fluctuations,  $\Delta T_i \sim \mathcal{O}(10)$  K. A wall thickness range of

$L \in [200, 300]$  mm was identified as optimal, minimizing the amplitude of these fluctuations while maintaining reasonable regolith mass/volume.

To further enhance thermal inertia and reduce temperature oscillations, we explored integrating the organic PCM n-hexadecane into the regolith wall in both single and multi-strand configurations. PCM effectiveness was shown to depend on the  $(\alpha/\varepsilon)$  ratio, with optimal performance —  $\Delta T_i \rightarrow 0$  K — achieved when the quantity of melted and solidified PCM remain balanced between repeated illumination and eclipse periods. Additionally, PCM integration would help mitigate long-term absorptivity degradation, providing a sustainable solution for stable habitat conditions ( $\Delta T_i \sim \mathcal{O}(1)$  K), and reduce the use of regolith. From a practical perspective, this entails a reduction in the required amount of binder, which is essential for the geopolymerization of regolith. Since both the binder and the PCM will likely need to be transported from Earth, this relationship suggests the existence of an optimal binder-to-PCM ratio that would minimize transported weight and volume, a critical consideration for sustainable lunar construction.

Overall, the present results support the use of coaxial 3D printing for lunar (and extraterrestrial) habitat construction. The internal wall structure, comprising the regolith shell and the PCM core, can be effectively controlled through printing parameters. This wall structure, together with the selection of an adequate, commercially available coating for the external boundary of the habitat, can achieve good thermal performance and stable living conditions in the interior.

Future research will incorporate more complex physics into the thermal model and extend this analysis to 2D and 3D cases. The effect of natural convection, for instance, will be introduced to describe this additional mechanism of heat transfer, while surface tension effects can be also retained to assess their influence on the internal wall structure and PCM dynamics. Specifically, PCM shrinkage after solidification may create voids before remelting, potentially affecting long-term performance. Although not directly analyzed here, these factors could prove critical for accurately predicting the durability and effectiveness of PCM-integrated habitat walls.

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**Data Availability** No datasets were generated or analysed during the current study.

## Declarations

**Competing Interests** Berin Šeta serves as an Associate Editor for Microgravity Science and Technology. The authors declare no other competing interests.

**Ethical Approval** Not applicable.

**Consent to Participate** Not applicable.

**Consent for Publication** Not applicable.

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